1M1B AI FOR SUSTAINABILITY SUBMISSION

Anna-Setu

AI-Powered Campus Food Rescue & Redistribution Network

Primary SDG: SDG 12 - Responsible Consumption and Production

Secondary SDG: SDG 2 - Zero Hunger

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1. Project Overview

Project Title: Anna-Setu - "Anna" (food/sustenance) + "Setu" (bridge) - the AI-powered bridge between campus surplus and community need.

Primary SDG Alignment: SDG 12 - Responsible Consumption and Production. The project directly targets Target 12.3, which calls for halving per-capita global food waste at the retail and consumer level by 2030.

Secondary SDG Alignment: SDG 2 - Zero Hunger. Rescued surplus is redirected to food-insecure individuals and community shelters, converting an institutional inefficiency into a source of local nutrition security.

Anna-Setu is a conceptual AI system for higher-education and school campuses that pairs predictive demand forecasting with an automated, policy-compliant surplus-matching pipeline turning cafeteria over-production from an environmental liability into a measurable social good.

2. Problem Statement

> ***"How might we use AI to accurately forecast daily meal demand and intelligently match unavoidable surplus food with verified local shelters, so that campus dining ecosystems and food-insecure communities can become more sustainable?"***

Contextualising the Problem

Most university and school mess halls and cafeterias operate on fixed, historical menu-planning cycles rather than on real, day-specific demand signals. This creates a structural mismatch between food prepared and food consumed:

- Overproduction as a default safety margin: Kitchen staff routinely over-cook by 10–30% to avoid the reputational and logistical cost of running out, with no data-driven basis for the margin chosen.
- No demand forecasting: Menus are planned weeks in advance on a rotating cycle, blind to variables that swing actual footfall, mid-semester and end-semester exam weeks, festival and holiday clusters, weather (heavy rain or extreme heat reduces walk-in attendance), and one-off campus events.
- Fragmented, manual surplus handling: When surplus does exist, there is typically no standing relationship, verified contact list, or safety-checked handoff process with a local shelter or food bank, so left-over food is either held unsafely, distributed ad hoc, or simply discarded.
- Waste ends in landfill, not redistribution: Without a forecasting-plus-matching system, the default outcome for surplus is disposal, which generates methane during anaerobic decomposition and wastes the water, energy, and labour embedded in producing that food in the first place.
- No feedback loop: Cafeteria managers rarely have visibility into how their over/under-production decisions compare to actual consumption over time, so the same inefficiencies repeat semester after semester.

These four gaps - no forecasting, no formal matching mechanism, no compliance layer, and no feedback loop - are precisely what an AI-driven system is positioned to close.

3. Detailed Solution Description (Design Thinking Approach)

Target Users

- Campus students and diners - as both demand-signal generators (via meal swipes, app RSVPs) and, where relevant, beneficiaries of reduced mess fees from cost savings.
- Cafeteria/mess management and kitchen staff - the primary decision-makers who plan procurement, cooking quantities, and staffing.
- Local food shelter coordinators and NGO partners - the receiving end of the redistribution pipeline, who need reliable, safe, and timely surplus notifications.
- Campus sustainability/administration office - who track SDG-linked impact metrics for institutional ESG and accreditation reporting.

Empathize & Define - Current Operational Gaps

- Kitchen staff have no simple, low-effort way to log daily consumption versus preparation quantities, so historical data (if it exists at all) is inconsistent or paper-based.
- Mess managers lack visibility into upcoming campus events, exam calendars, or weather forecasts at the point of menu and quantity planning.
- There is no trusted, pre-vetted directory of shelters willing and able to receive surplus at short notice, nor a standard protocol for what is safe to donate.
- Students and diners have no channel to signal attendance intent in advance (e.g., “skipping dinner tonight”), so demand remains a guess rather than a forecast.
- Sustainability data - waste weight, cost of over-production, meals redirected - is not centrally tracked, so the institution cannot demonstrate or improve its SDG 12 performance.

Ideation - The Solution: Smart Campus Dining Advisor & Predictive Redistribution Pipeline

Anna-Setu is built around two connected AI-driven modules:

Module 1 - Smart Campus Dining Advisor: A predictive analytics layer that forecasts next-day (and next-meal) attendance and consumption per menu item, using historical mess logs, the academic calendar (exams, holidays, events), and weather data. It issues a recommended cooking quantity to kitchen staff each morning, with a stated confidence range, replacing the current “cook for a fixed number, hope for the best” approach.

Module 2 - Predictive Redistribution Pipeline: An automated matching and compliance layer that activates when the Advisor detects (or kitchen staff report) a surplus above a safe-to-donate threshold. It checks the surplus against food-safety and holding-time rules, identifies the nearest available and verified shelter partner from a maintained directory, and drafts and sends a structured surplus alert, by text or WhatsApp-style message, so a pickup or drop-off can be arranged within a safe time window.

Together, the two modules shift the campus dining operation from reactive waste disposal to proactive waste prevention plus responsible redistribution of whatever surplus remains unavoidable.

Expected Impact

Dimension	Expected Impact
Economic	Reduced cafeteria food-procurement costs through demand-matched cooking (targeted 15–25% reduction in over-production); lower institutional waste-disposal fees; potential CSR/ESG recognition for the campus.
Environmental	Diverted food waste no longer decomposes anaerobically in landfills, cutting methane emissions; lower water, energy, and land footprint associated with wasted food production; supports the institution's sustainability reporting.

Social	Consistent, dignified access to surplus meals for local shelters, day-labourers, and food-insecure families; strengthened campus-community partnerships; measurable contribution to zero-hunger goals in the surrounding area.

4. AI Elements, Tools, & Logical Workflow

Core AI Components

Predictive Analytics / Machine Learning: A time-series and regression-based forecasting model (e.g., gradient-boosted trees or a simple LSTM) trained on historical meal-log data, day-of-week and academic-calendar features (exam periods, holidays, orientation weeks), and external weather data. Output: a per-meal, per-menu-item demand forecast with a confidence interval, refreshed daily.

IBM Granite Models & Prompt Engineering: An IBM Granite foundation model, driven by structured prompt templates, powers the natural-language layer of the system - turning structured surplus data (quantity, item, safe-until time, pickup location) into a clear, actionable WhatsApp/SMS-style broadcast message for shelter coordinators, and answering kitchen-staff questions about the day's forecast in plain language.

Retrieval-Augmented Generation (RAG): A RAG pipeline indexes local municipal food-safety guidelines, FSSAI (or equivalent regional) food-handling and donation-compliance documents, and the institution's own donation policy. Before any surplus alert is generated, the system retrieves the relevant compliance clauses so the drafted message and the underlying recommendation stay grounded in actual, current regulation rather than the model's unguided assumptions.

Step-by-Step Logical Workflow

Step	Stage	Description
1	**Data Collection**	Daily meal-swipe counts, kitchen preparation quantities, and plate-waste logs are recorded (manually via a simple form, or automatically via POS/turnstile data). Academic-calendar events and next-day weather are pulled in automatically.
2	**Demand Forecasting**	The ML model processes the combined dataset each evening and produces the next day's recommended cooking quantity per menu item, with a confidence range and a plain-language explanation of the key drivers.
3	**Kitchen Decision Support**	Mess managers view the forecast on a simple dashboard, can accept or manually adjust it, and place their procurement/cooking order for the next day — reducing the default over-production margin.
4	**Real-Time Surplus Detection**	During and after service, the system compares actual consumption against quantity prepared. If leftover food exceeds a configurable safe-to-donate threshold, it flags a surplus event.
5	**Compliance Check (RAG)**	The RAG pipeline retrieves the applicable food-safety and holding-time rules for the flagged food category and confirms the surplus is within a safe donation window before proceeding.
6	**Shelter Matching**	The system checks a maintained directory of verified local shelters/NGOs for availability, proximity, and capacity, and selects the best-matched recipient(s) for the quantity and type of surplus.
7	**AI-Drafted Alert (Granite + Prompting)**	The IBM Granite model, using the prompt template in Section 5, converts the structured surplus and compliance data into a clear, human-readable WhatsApp/SMS broadcast to the matched shelter coordinator(s).
8	**Human-in-the-Loop Confirmation**	A designated kitchen/administration staff member reviews and approves the drafted alert before it is sent, keeping a human decision-maker in control of the final action.
9	**Pickup / Redistribution**	The shelter coordinator confirms and arranges pickup or drop-off within the specified safe window; the handoff is logged for traceability.

10	**Feedback & Reporting**	Actual outcomes (quantity redistributed, forecast accuracy, cost saved, estimated methane avoided) are logged and fed back into the model for continuous improvement, and rolled up into a sustainability dashboard for the campus.

5. Prototype Logic & Prompt Samples

The example below shows a system prompt and structured input that would drive an LLM (e.g., an IBM Granite model) to automatically draft a compliant, ready-to-send "Surplus Food Alert" broadcast once Module 2 has confirmed a safe, matched surplus event.

System Prompt

```
SYSTEM PROMPT:
You are the messaging assistant for Anna-Setu, a campus food-redistribution
system. Your only task is to convert structured surplus-food data into a
short, clear, and warm broadcast message for a verified shelter coordinator.

Rules you must always follow:
1. Use only the facts provided in the input data. Never invent quantities,
   times, food items, or locations.
2. Always state the safe-pickup-by time exactly as given.
3. Reference the retrieved compliance note in one short line so the
   coordinator knows the food has been checked against safety guidelines.
4. Keep the message under 60 words, in plain, respectful language suitable
   for WhatsApp/SMS.
5. End with a clear call to action (how to confirm pickup).
6. Do not include any student-identifying or personal data.
```

Structured Input (Retrieved + Detected Data)

```
{
  "surplus_item": "Vegetable pulao and dal",
  "quantity_servings": 45,
  "prepared_time": "13:05",
  "safe_pickup_by": "15:30",
  "pickup_location": "Main Campus Mess, Gate 2",
  "compliance_note": "Within 3-hour ambient holding limit per municipal",
  "matched_shelter": "Asha Community Shelter (2.1 km away, confirmed capacity)",
  "contact_channel": "WhatsApp Business number"
}
```

Expected Model Output (Draft Broadcast)

```
🍽 Surplus Food Alert — Anna-Setu
```

```
Hi Asha Community Shelter team, we have 45 servings of vegetable pulao
and dal ready for pickup, prepared at 1:05 PM and safe until 3:30 PM
(within the municipal 3-hour holding guideline).

Pickup point: Main Campus Mess, Gate 2.

Please reply YES to confirm pickup, or call us if you need a later
window. Thank you for partnering with us!
```

This same prompt-chain pattern can be extended with a follow-up prompt that logs the coordinator's confirmation and updates the redistribution record automatically.

6. Responsible AI Considerations (Mandatory)

Anna-Setu is designed with responsible AI as a first-class requirement, not an afterthought. The table below maps each mandated pillar to a concrete design decision.

Pillar	Focus	How Anna-Setu Addresses It
Fairness	*Avoid biased assumptions across demographics, hostel blocks, or meal preferences*	Forecasting models are trained on pooled, de-identified consumption logs rather than individually profiled students; demand predictions are validated across dietary categories (vegetarian, non-vegetarian, Jain, vegan) and residential blocks to ensure no group is systematically under- or over-served. Model outputs are periodically audited against actual consumption to catch drift or skew.
Transparency	*Explainable predictions for cafeteria managers and administrators*	Every demand forecast is accompanied by a plain-language rationale (e.g., 'Predicted 12% lower footfall - mid-semester exams begin tomorrow and historical logs show a similar dip'). Dashboards expose the top three factors driving each prediction so non-technical mess staff can question or override recommendations.
Ethics	*Strict food safety and compliance adherence*	The system will never recommend redistribution of food that has exceeded safe holding-temperature windows or hygiene thresholds. All redistribution suggestions are gated by a rules layer that checks preparation time, storage conditions, and locally applicable food-safety guidelines before any alert is generated.
Privacy	*Protecting student data anonymity*	Meal-swipe and consumption data are aggregated at the cohort level (e.g., 'Block C dinner turnout') and stripped of individual identifiers before being used for forecasting. No student-level eating pattern, health condition, or dietary restriction is ever shared with external shelter partners or stored beyond the academic term.

7. Submission Placeholders

Asset	Placeholder
GitHub Repository Link	https://github.com/pgovila/Anna-Setu-Project
Live Demo / Prototype Link	https://huggingface.co/spaces/pgovila/Anna-Setu-Demo
Demo Video Link	https://drive.google.com/drive/folders/1089iQX6AttZPXIuL8e6_bv6nGzkhyZW3?usp=sharing
Team / Presenter Details	*Pankhuri Govila | University of Petroleum and Energy Sources UPES*

Closing Note: Disclaimer

The prototype is under development and the report is prepared on the basis of initial study analysis and deployment, it is subject to change in the coming months.